Safety Code 6 (SC6) in Plain English

What is Safety Code 6?
Safety Code 6 is Health Canada's standard for limiting human exposure to radio frequency (RF) energy. It applies to cell towers, 4G/5G networks, Wi-Fi, radio transmitters, microwave links, and public safety systems. Its purpose is to make sure RF exposure stays below safe limits for workers and the public.

What Does an SC6 Analyst Do?
An SC6 analyst reviews wireless sites, checks antenna and RF exposure data, confirms compliance, identifies issues, and recommends solutions such as signage, barriers, access restrictions, antenna adjustments, or power reductions.

RF Basics
RF (Radio Frequency) is electromagnetic energy used for wireless communication. RF is a type of EMF (Electromagnetic Field). RF radiation is non-ionising, meaning it does not have enough energy to ionise atoms like X-rays or gamma rays.

Important Antenna Terms
Gain: How well an antenna focuses energy.
Azimuth: The direction an antenna faces.
Mechanical Tilt: Physical antenna angle.
Electrical Tilt: Electronic adjustment of the antenna beam.
EIRP: Effective radiated power after considering gain and losses.

Power Density
Measures RF energy over an area (W/m^2). Results are often shown as a percentage of the SC6 limit. 100% means the exposure is at the allowable limit.

SAR
Specific Absorption Rate (W/kg) measures RF energy absorbed by the body.

Controlled vs Uncontrolled Areas
Uncontrolled areas are public spaces and have stricter limits. Controlled areas are restricted to trained workers and may allow higher occupational exposure limits.

Near Field vs Far Field
Near field is close to the antenna and has complex RF behaviour. Far field is farther away and easier to predict and calculate.

Multiple RF Sources
SC6 evaluates combined exposure from all transmitters at a site, not just one source.

Typical Assessment Process
Collect site data -> Review antenna data -> Perform RF modelling -> Calculate exposure -> Compare to limits -> Determine compliance -> Recommend actions -> Issue report.

Common Compliance Issues
Public exposure above limits, worker exposure concerns, and rooftop exposure issues. Typical solutions include barriers, signage, restricted access, antenna changes, and power reductions.

RF Safety Signs
Notice: RF equipment present.
Warning: Elevated RF exposure may exist.
Danger: Exposure could exceed safe limits.

Key Formula

Power Density = S = P / A

Common Software
EMF Visual, iBwave, EDX SignalPro, and Atoll are commonly used to model RF exposure and support compliance assessments.

Summary
Safety Code 6 is Canada's RF exposure standard. SC6 assessments ensure RF exposure from wireless systems remains below allowable limits for workers and the public.